

Introduction to Machine Learning

Lecture 4: Principal Component Analysis

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Agenda for This Week: *Dimensionality Reduction*

In today's lecture, we study our last *unsupervised learning* task, i.e.,
dimensionality reduction

through the following steps

- *Representing data in lower dimension*
 - ↳ *We review key notions in linear algebra*
- *Principal component analysis*
- *A look at particular applications*
 - ↳ *Image compression*
 - ↳ *Recommendation systems*

Motivation: *Compression*

Say we have data samples in D -dimensional space

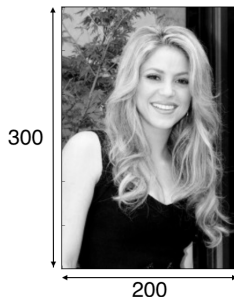
$$\mathbb{D} = \{\mathbf{x}_n \in \mathbb{R}^D : n = 1, \dots, N\}$$

? *Can we represent it in lower dimension?*

$$f : \mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_D \end{bmatrix} \mapsto \mathbf{z} = \begin{bmatrix} z_1 \\ \vdots \\ z_K \end{bmatrix}$$

Example: Image Compression

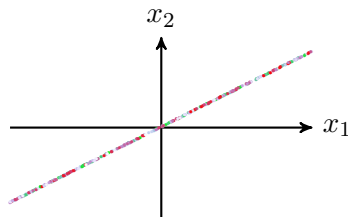
Consider a 60K pixel image



? *Can we represent it with much less pixel while maintaining quality?*

Representing Data in Lower Dimensions: *Example 1*

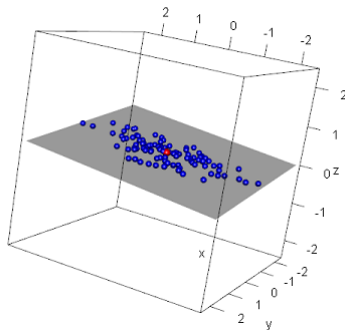
Consider the example where all samples lie on a line



! Obviously, we can represent each sample with only one scalar!

Representing Data in Lower Dimensions: *Example II*

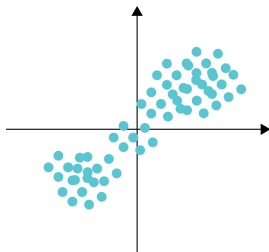
Consider the example where all samples lie on a 2D plane



! We can represent each sample with a 2D vector

Representing Data in Lower Dimensions: *General Form*

In reality, the samples might be more spread in one direction



! *We can have an approximate low-dimensional representation*

Orthonormal Bases

$\mathbf{u}_i \in \mathbb{R}^D$ form an orthonormal base if

$$\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_D] \rightsquigarrow \begin{cases} \mathbf{u}_j^\top \mathbf{u}_j = 1 \\ \mathbf{u}_i^\top \mathbf{u}_j = 0 \end{cases} \rightsquigarrow \mathbf{U}^\top \mathbf{U} = \mathbf{I}_D$$

Classic Base

Classic base is given by the identity matrix $\mathbf{I}_D$

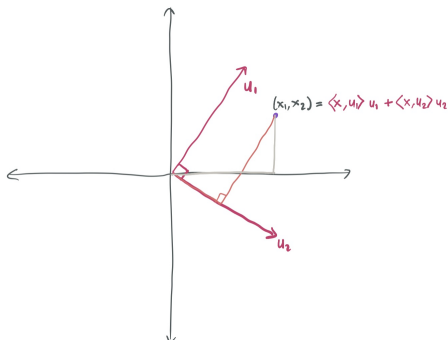
Key Feature

Orthonormal base only rotates and does not change the norm

$$\|\mathbf{U}\mathbf{x}\| = \|\mathbf{x}\|$$

Orthonormal Bases

We can use orthonormal bases to represent a vector



Recall

In D -dimensional space, we have only D orthogonal vectors

Eigenvalues and Eigenvectors

Say we have a square matrix $\mathbf{A} \in \mathbb{R}^{D \times D}$

Eigenvalue and Eigenvector

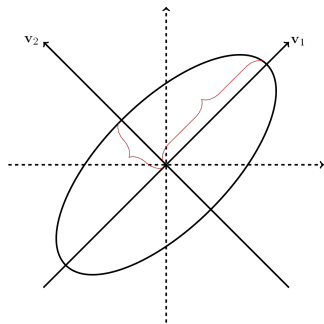
$(\lambda, \mathbf{v})$ is an eigenvalue and eigenvector pair if

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

Recall

We consider eigenvectors to be unit-norm, i.e., $\|\mathbf{v}\| = 1$

Eigenvalues and Eigenvectors: *Visualization*



! *Eigenvectors describe an orthonormal base*

Matrix Decomposition

Let's put everything in a matrix form

$$\begin{aligned}\mathbf{A} [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_D] &= [\lambda_1 \mathbf{v}_1 \quad \cdots \quad \lambda_D \mathbf{v}_D] \\ &= [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_D] \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_D \end{bmatrix}\end{aligned}$$

Eigendecomposition

We can hence say that

$$\mathbf{A} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top$$

where $\mathbf{V}$ is an orthonormal base and $\mathbf{\Lambda}$ is diagonal matrix of eigenvalues

Key Features

- We can compute determinant from eigenvalues

$$\det \mathbf{A} = \prod_{i=1}^D \lambda_i$$

- We can compute trace from eigenvalues

$$\text{tr}\{\mathbf{A}\} = \sum_{i=1}^D \lambda_i$$

- A *positive semi-definite* matrix has only non-negative eigenvalues

$$\mathbf{A} \geq 0 \rightsquigarrow \lambda_i \geq 0$$

↳ Famous example of a positive semi-definite matrix

$$\mathbf{A} = \mathbf{X}\mathbf{X}^T$$

Dimensionality Reduction: *Problem Formulation*

Single direction $\mathbf{u} \in \mathbb{R}^D$

$$\hat{\mathbf{x}}_n = z_n \mathbf{u} + \boldsymbol{\mu}$$

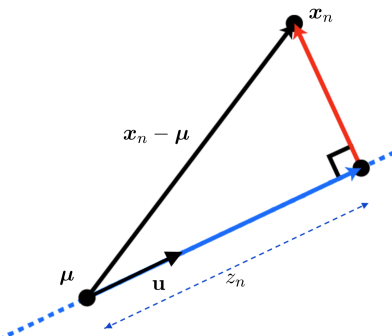
Of course $\mathbf{u}$ is a base, i.e.,

$$\|\mathbf{u}\| = 1$$

Dimensionality Reduction: *Problem Formulation*

How to find z_n ?

$$z_n = \mathbf{u}^T (\mathbf{x}_n - \boldsymbol{\mu})$$



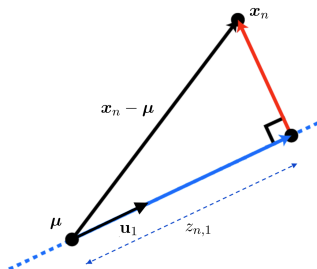
Dimensionality Reduction: *Problem Formulation*

Two directions $\mathbf{u}_1$ and $\mathbf{u}_2 \in \mathbb{R}^D$

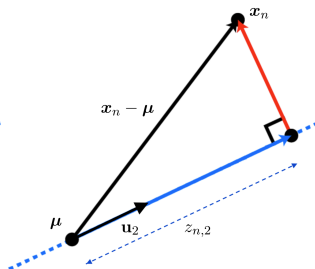
$$\begin{aligned}\hat{\mathbf{x}}_n &= z_{n,1}\mathbf{u}_1 + z_{n,2}\mathbf{u}_2 + \boldsymbol{\mu} \\ &= [\mathbf{u}_1, \mathbf{u}_2] \begin{bmatrix} z_{n,1} \\ z_{n,2} \end{bmatrix} + \boldsymbol{\mu} = \mathbf{U}\mathbf{z}_n + \boldsymbol{\mu}\end{aligned}$$

How to find z_n ?

$$z_{n,1} = \mathbf{u}_1^\top (\mathbf{x}_n - \boldsymbol{\mu})$$



$$z_{n,2} = \mathbf{u}_2^\top (\mathbf{x}_n - \boldsymbol{\mu})$$



Dimensionality Reduction: *Problem Formulation*

So we can write

$$\begin{aligned} \mathbf{z}_n = \begin{bmatrix} z_{n,1} \\ z_{n,2} \end{bmatrix} &= \begin{bmatrix} \mathbf{u}_1^\top (\mathbf{x}_n - \boldsymbol{\mu}) \\ \mathbf{u}_2^\top (\mathbf{x}_n - \boldsymbol{\mu}) \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1^\top \\ \mathbf{u}_2^\top \end{bmatrix} (\mathbf{x}_n - \boldsymbol{\mu}) \\ &= [\mathbf{u}_1, \mathbf{u}_2]^\top (\mathbf{x}_n - \boldsymbol{\mu}) \\ &= \mathbf{U}^\top (\mathbf{x}_n - \boldsymbol{\mu}) \end{aligned}$$

Dimensionality Reduction: *Problem Formulation*

$\mathbf{u}_1$ and $\mathbf{u}_2$ are bases, i.e.,

$$\|\mathbf{u}_1\| = \|\mathbf{u}_2\| = 1 \qquad \mathbf{u}_1^T \mathbf{u}_2 = 0$$

So we can say

$$\mathbf{U}^T \mathbf{U} = \begin{bmatrix} \|\mathbf{u}_1\|^2 & \mathbf{u}_1^T \mathbf{u}_2 \\ \mathbf{u}_1^T \mathbf{u}_2 & \|\mathbf{u}_2\|^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}_2$$

Orthonormal bases!

Dimensionality Reduction: *Problem Formulation*

$K < D$ directions $\mathbf{u}_1, \dots, \mathbf{u}_K \in \mathbb{R}^D$

$$\begin{aligned}\hat{\mathbf{x}}_n &= \sum_{k=1}^K z_{n,k} \mathbf{u}_k + \boldsymbol{\mu} \\ &= [\mathbf{u}_1, \dots, \mathbf{u}_K] \begin{bmatrix} z_{n,1} \\ \vdots \\ z_{n,K} \end{bmatrix} + \boldsymbol{\mu} \\ &= \mathbf{U} \mathbf{z}_n + \boldsymbol{\mu}\end{aligned}$$

Similarly, we can find

$$\mathbf{z}_n = \mathbf{U}^T (\mathbf{x}_n - \boldsymbol{\mu})$$

Dimensionality Reduction: *Problem Formulation*

We know that $\mathbf{U}$ contains orthonormal bases

$$\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_K] \rightsquigarrow \begin{cases} \mathbf{u}_k^\top \mathbf{u}_j = 0 \\ \|\mathbf{u}_k\| = 1 \end{cases} \rightsquigarrow \mathbf{U}^\top \mathbf{U} = \mathbf{I}_K$$

Attention

Note that $\mathbf{U} \in \mathbb{R}^{D \times K}$ and that

$$\mathbf{U}\mathbf{U}^\top \neq \mathbf{I}_D$$

Latent Space $\longleftrightarrow$ Reconstruction

Dimensionality Reduction: *Latent Variable*

Dimensionality is reduced linearly via $\mathbf{U}^T : \mathbb{R}^D \mapsto \mathbb{R}^K$ as

$$\mathbf{z}_n = \mathbf{U}^T (\mathbf{x}_n - \boldsymbol{\mu})$$

We call $\mathbf{z}_n$ *latent* variable

Higher Dimensional Recovery: *Reconstruction*

We reconstruct a sample $\mathbf{x}_n \in \mathbb{R}^D$ from its latent variable $\mathbf{z}_n \in \mathbb{R}^K$ as

$$\hat{\mathbf{x}}_n = \mathbf{U} \mathbf{z}_n + \boldsymbol{\mu}$$

Dimensionality Reduction as *Learning Task*

- Data

- ↳ Collection of samples $\mathbb{D} = \{\mathbf{x}_n : n = 1, \dots, N\}$

- Model

- ↳ Linear transform $\mathbf{z} = \mathbf{U}^T (\mathbf{x} - \boldsymbol{\mu})$

- Learning algorithm

- ↳ We look for algorithm $\mathcal{A} : \mathbb{D} \mapsto \mathbf{U}^*, \boldsymbol{\mu}^*$

- ↳ $\mathbf{U}^*, \boldsymbol{\mu}^*$ are **good** ones!

? *How can we define a **good** transform?*

! *It should not kill to much information!*

Preserving Information Maximally

Say we specify $\mathbf{U}$ and $\boldsymbol{\mu}$ and compute **latent variable** $\mathbf{z}$ of sample $\mathbf{x}$: if we want the data sample back, we reconstruct from the latent space as

$$\hat{\mathbf{x}} = \mathbf{U}\mathbf{z} + \boldsymbol{\mu}$$

The information on $\mathbf{x}_n$ is preserved if

$$\hat{\mathbf{x}} \stackrel{!}{=} \mathbf{x} \iff \|\hat{\mathbf{x}} - \mathbf{x}\|^2 \stackrel{!}{=} 0$$

But, we know that it's not happening perfectly; thus, we try

$$\|\hat{\mathbf{x}} - \mathbf{x}\|^2 \stackrel{!}{\approx} 0$$

? We cannot check it for every data sample!

! But, we can check it for samples in $\mathbb{D}$!

Information Preservation $\equiv$ Minimum Representation Error

We want $\mathbf{U}$ and $\boldsymbol{\mu}$ to make $\|\hat{\mathbf{x}}_n - \mathbf{x}_n\|^2$ as small as possible for $\mathbf{x}_n \in \mathbb{D}$: so we could try to make the average

$$\mathcal{J}(\mathbf{U}, \boldsymbol{\mu}) = \frac{1}{N} \sum_{n=1}^N \|\hat{\mathbf{x}}_n - \mathbf{x}_n\|^2$$

is minimized $\rightsquigarrow$ if $\mathcal{J}(\mathbf{U}, \boldsymbol{\mu}) = 0$; then, $\hat{\mathbf{x}}_n = \mathbf{x}_n$ for all n

Dimensionality Reduction via Minimum Representation Error

Optimal $\mathbf{U}^*$ and $\boldsymbol{\mu}^*$ are defined as

$$\mathbf{U}^*, \boldsymbol{\mu}^* = \underset{\mathbf{U}, \boldsymbol{\mu}}{\operatorname{argmin}} \mathcal{J}(\mathbf{U}, \boldsymbol{\mu})$$

Optimal Bias $\equiv$ Data Centroid

Let's start with simpler one: we want to find μ^*

$$\begin{aligned}\mathcal{J} &= \frac{1}{N} \sum_{n=1}^N \|\hat{\mathbf{x}}_n - \mathbf{x}_n\|^2 = \frac{1}{N} \sum_{n=1}^N \|\mathbf{U}\mathbf{z}_n + \boldsymbol{\mu} - \mathbf{x}_n\|^2 \\ &= \frac{1}{N} \sum_{n=1}^N \|\mathbf{U}\mathbf{U}^T (\mathbf{x}_n - \boldsymbol{\mu}) + \boldsymbol{\mu} - \mathbf{x}_n\|^2 \\ &= \frac{1}{N} \sum_{n=1}^N \|\left(\mathbf{U}\mathbf{U}^T - \mathbf{I}_D\right) (\mathbf{x}_n - \boldsymbol{\mu})\|^2\end{aligned}$$

Let's call $\mathbf{A} = \mathbf{U}\mathbf{U}^T - \mathbf{I}_D$: we want to find μ^* which minimizes

$$\mathcal{J} = \frac{1}{N} \sum_{n=1}^N \|\mathbf{A}\mathbf{x}_n - \mathbf{A}\boldsymbol{\mu}\|^2$$

Optimal Bias $\equiv$ Data Centroid

If we call $\mathbf{y}_n = \mathbf{A}\mathbf{x}_n$ and $\mathbf{a} = \mathbf{A}\boldsymbol{\mu}$; then, we look for

$$\mathbf{a}^* = \underset{\mathbf{a}}{\operatorname{argmin}} \frac{1}{N} \sum_{n=1}^N \|\mathbf{y}_n - \mathbf{a}\|^2$$

which is its centroid, i.e.,

$$\mathbf{a}^* = \frac{1}{N} \sum_{n=1}^N \mathbf{y}_n$$

So, we have

$$\mathbf{A}\boldsymbol{\mu}^* = \frac{1}{N} \sum_{n=1}^N \mathbf{A}\mathbf{x}_n \rightsquigarrow \boldsymbol{\mu}^* = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$$

Alternative Formulation

Let's now have a few observations: *call the error of sample n*

$$\mathbf{e}_n = \hat{\mathbf{x}}_n - \mathbf{x}_n$$

Note that our objective is

$$\mathcal{J}(\mathbf{U}, \boldsymbol{\mu}) = \frac{1}{N} \sum_{n=1}^N \|\hat{\mathbf{x}}_n - \mathbf{x}_n\|^2 = \frac{1}{N} \sum_{n=1}^N \|\mathbf{e}_n\|^2$$

We know that $\hat{\mathbf{x}}_n = \mathbf{U}\mathbf{z}_n + \boldsymbol{\mu}^*$, so we can write

$$\begin{aligned}\mathbf{e}_n &= \mathbf{U}\mathbf{z}_n + \boldsymbol{\mu}^* - \mathbf{x}_n \\ &= \mathbf{U}\mathbf{z}_n - (\mathbf{x}_n - \boldsymbol{\mu}^*)\end{aligned}$$

Alternative Formulation

Now, let's compute the inner product of $\mathbf{e}_n$ and $\hat{\mathbf{x}}_n - \boldsymbol{\mu}^*$, i.e.,

$$\phi_n = (\hat{\mathbf{x}}_n - \boldsymbol{\mu}^*)^\top \mathbf{e}_n$$

Recall that $\hat{\mathbf{x}}_n = \mathbf{U}\mathbf{z}_n + \boldsymbol{\mu}^*$, so we can write

$$\phi_n = (\hat{\mathbf{x}}_n - \boldsymbol{\mu}^*)^\top \mathbf{e}_n = \mathbf{z}_n^\top \mathbf{U}^\top \mathbf{e}_n$$

Alternative Formulation

So, the inner product reads

$$\begin{aligned}
 \phi_n &= \mathbf{z}_n^T \mathbf{U}^T \mathbf{e}_n \\
 &= \mathbf{z}_n^T \mathbf{U}^T (\mathbf{U} \mathbf{z}_n - (\mathbf{x}_n - \boldsymbol{\mu}^*)) \\
 &= \mathbf{z}_n^T \mathbf{U}^T \mathbf{U} \mathbf{z}_n - \mathbf{z}_n^T \mathbf{U}^T (\mathbf{x}_n - \boldsymbol{\mu}^*)
 \end{aligned}$$

Recall that $\mathbf{U}^T \mathbf{U} = \mathbf{I}_K$, so we can write

$$\mathbf{z}_n^T \mathbf{U}^T \mathbf{U} \mathbf{z}_n = \mathbf{z}_n^T \mathbf{z}_n$$

Also we know that $\mathbf{U}^T (\mathbf{x}_n - \boldsymbol{\mu}^*) = \mathbf{z}_n$, so we could say

$$\mathbf{z}_n^T \mathbf{U}^T (\mathbf{x}_n - \boldsymbol{\mu}^*) = \mathbf{z}_n^T \mathbf{z}_n$$

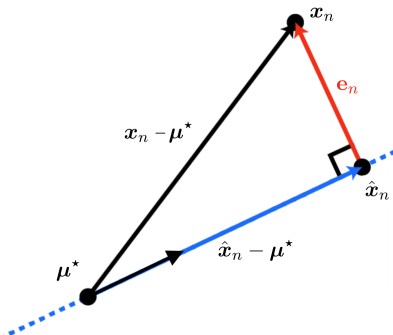
This means that

$$\phi_n = \mathbf{z}_n^T \mathbf{z}_n - \mathbf{z}_n^T \mathbf{z}_n = 0$$

Alternative Formulation

Orthogonality Principle

The **error** $\mathbf{e}_n$ and **unbiased estimate** $\hat{\mathbf{x}}_n - \boldsymbol{\mu}^*$ are orthogonal



We can hence write

$$\|\hat{\mathbf{x}}_n - \boldsymbol{\mu}^*\|^2 + \|\mathbf{e}_n\|^2 = \|\mathbf{x}_n - \boldsymbol{\mu}^*\|^2$$

Alternative Formulation

We have seen that

$$\|\hat{\mathbf{x}}_n - \boldsymbol{\mu}^\star\|^2 = \|\mathbf{U}\mathbf{z}_n\|^2 = \|\mathbf{z}_n\|^2$$

and also know that $\|\mathbf{x}_n - \boldsymbol{\mu}^\star\|^2$ has nothing to do with $\mathbf{U}$; thus,

$$\|\mathbf{z}_n\|^2 + \|\mathbf{e}_n\|^2 = \text{constant}$$

If we average over n , we conclude

$$\frac{1}{N} \sum_{n=1}^N \|\mathbf{z}_n\|^2 + \mathcal{J} = \text{constant} \rightsquigarrow \mathcal{J} = \text{constant} - \frac{1}{N} \sum_{n=1}^N \|\mathbf{z}_n\|^2$$

Alternative Formulation: *Maximal Representation Variance*

Dimensionality Reduction via *Minimum Representation Error*

Optimal $\mathbf{U}^*$ is given by

$$\mathbf{U}^* = \underset{\mathbf{U}}{\operatorname{argmin}} \mathcal{J}(\mathbf{U}, \boldsymbol{\mu})$$

Dimensionality Reduction via *Maximal Representation Variance*

Optimal $\mathbf{U}^*$ is given by

$$\mathbf{U}^* = \underset{\mathbf{U}}{\operatorname{argmax}} \frac{1}{N} \sum_{n=1}^N \|z_n\|^2$$

Notion of Correlation in Data

? *Why do we call it variance?*

We know that $\mathbf{U}^\top (\mathbf{x}_n - \boldsymbol{\mu}^\star) = \mathbf{z}_n$, so we can say

$$\begin{aligned}\|\mathbf{z}_n\|^2 &= \|\mathbf{U}^\top \mathbf{x}_n - \mathbf{U}^\top \boldsymbol{\mu}^\star\|^2 = \|\mathbf{U}^\top \mathbf{x}_n - \frac{1}{N} \sum_{n=1}^N \mathbf{U}^\top \mathbf{x}_n\|^2 \\ &= \|\mathbf{U}^\top \mathbf{x}_n - \text{avg}(\mathbf{U}^\top \mathbb{D})\|^2\end{aligned}$$

Therefore, we could say

$$\begin{aligned}\frac{1}{N} \sum_{n=1}^N \|\mathbf{z}_n\|^2 &= \text{var}(\mathbf{U}^\top \mathbb{D}) \\ &= \text{variance in the latent space}\end{aligned}$$

Notion of Variance in Data

? *How can we maximize the variance?*

Say $K = 1$, i.e., $\mathbf{U} = \mathbf{u} \rightsquigarrow z_n = \mathbf{u}^\top (\mathbf{x}_n - \boldsymbol{\mu}^\star)$

$$\begin{aligned}\frac{1}{N} \sum_{n=1}^N |z_n|^2 &= \frac{1}{N} \sum_{n=1}^N |\mathbf{u}^\top (\mathbf{x}_n - \boldsymbol{\mu}^\star)|^2 \\ &= \frac{1}{N} \|\mathbf{u}^\top \tilde{\mathbf{X}}\|^2\end{aligned}$$

where we define

$$\tilde{\mathbf{X}} = [\mathbf{x}_1 - \boldsymbol{\mu}^\star, \dots, \mathbf{x}_N - \boldsymbol{\mu}^\star] \in \mathbb{R}^{D \times N}$$

Sample Covariance

Sample Covariance

The sample covariance of the dataset is given by

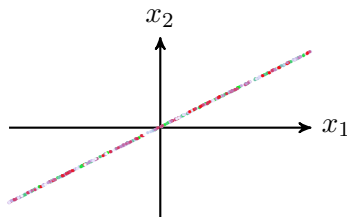
$$\Sigma = \frac{1}{N} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^T$$

In case of $K = 1$, we can write the variance as

$$\begin{aligned} \frac{1}{N} \sum_{n=1}^N |z_n|^2 &= \frac{1}{N} \mathbf{u}^T \tilde{\mathbf{X}} \tilde{\mathbf{X}}^T \mathbf{u} \\ &= \mathbf{u}^T \Sigma \mathbf{u} \end{aligned}$$

Sample Covariance

Let's compute sample covariance for the line example

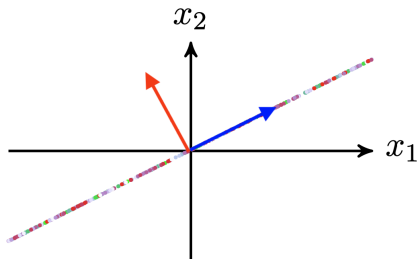


$$\tilde{\mathbf{X}} = \begin{bmatrix} -2 & -1 & 0 & 1 & 2 \\ -2 & -1 & 0 & 1 & 2 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 20 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

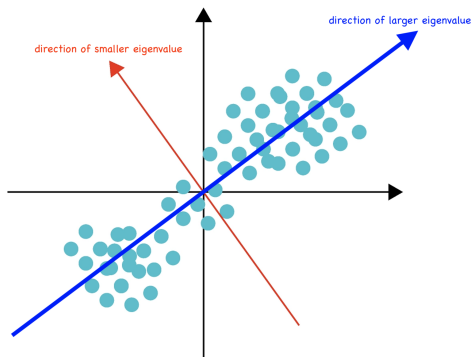
Sample Covariance

The large eigenvalue shows us where the data spans more



Sample Covariance

This applies to any dataset



! *It can tell us right direction for dimensionality reduction!*

Optimal Transform $\equiv$ *Principal Components of Covariance*

Σ is positive semi-definite: we thus have

$$\Sigma = [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_D] \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_D \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_D^\top \end{bmatrix}$$

with $\lambda_1, \dots, \lambda_D \geq 0$. Let's replace in the variance expression

$$\begin{aligned} \frac{1}{N} \sum_{n=1}^N |z_n|^2 &= \mathbf{u}^\top \Sigma \mathbf{u} \\ &= \mathbf{u}^\top [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_D] \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_D \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_D^\top \end{bmatrix} \mathbf{u} \\ &= \sum_{i=1}^D \lambda_i \left(\mathbf{u}^\top \mathbf{v}_i \right)^2 \end{aligned}$$

Optimal Transform $\equiv$ *Principal Components of Covariance*

We want to find optimal $\mathbf{u}$: *it should maximize the following term*

$$\frac{1}{N} \sum_{n=1}^N |z_n|^2 = \sum_{i=1}^D \lambda_i \left(\mathbf{u}^\top \mathbf{v}_i \right)^2$$

- $\mathbf{v}_i$'s are bases; thus, we always have

$$\left(\mathbf{u}^\top \mathbf{v}_i \right)^2 \leq 1$$

- $[\mathbf{v}_1 \cdots \mathbf{v}_D]$ is an orthonormal matrix; thus,

$$\|\mathbf{u}^\top [\mathbf{v}_1 \cdots \mathbf{v}_D]\| = \sum_{i=1}^D \left(\mathbf{u}^\top \mathbf{v}_i \right)^2 = 1$$

Optimal Transform $\equiv$ *Principal Components of Covariance*

We want to find optimal $\mathbf{u}$: *its should be set to*

$$\mathbf{u} = \mathbf{v}_{\max}$$

where $\mathbf{v}_{\max}$ corresponds to the maximum eigenvalue $\lambda_{\max}$

$$\frac{1}{N} \sum_{n=1}^N |z_n|^2 = \lambda_{\max}$$

Optimal Transform $\equiv$ *Principal Components of Covariance*

Say $K = 2$, i.e., $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2] \rightsquigarrow z_{n,1}, z_{n,2}$

$$\begin{aligned}\frac{1}{N} \sum_{n=1}^N \|\mathbf{z}_n\|^2 &= \frac{1}{N} \sum_{n=1}^N |z_{n,1}|^2 + \frac{1}{N} \sum_{n=1}^N |z_{n,2}|^2 \\ &= \mathbf{u}_1^T \Sigma \mathbf{u}_1 + \mathbf{u}_2^T \Sigma \mathbf{u}_2\end{aligned}$$

With same lines of derivations, we could say

- Optimal $\mathbf{u}_1$ is $\mathbf{v}_{\max_1}$ corresponding to the largest eigenvalue
- Optimal $\mathbf{u}_2$ is $\mathbf{v}_{\max_2}$ corresponding to the second largest eigenvalue

Principal Component Analysis: *General Algorithm*

PCA() :

1: Set μ to the centroid of dataset

$$\mu = \frac{1}{N} \sum_{n=1}^N x_n$$

2: Construct the sample covariance matrix

$$\Sigma = \frac{1}{N} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^T$$

3: Decompose Σ in terms of its eigenvalues and eigenvectors as $\Sigma = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$

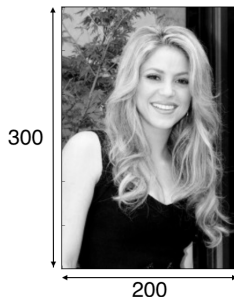
4: Find the K largest eigenvalues and set $\mathbf{u}_k = \mathbf{v}_{\max_k}$

5: Set the projection matrix to

$$\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_K]$$

Formulating the Problem

We can look at an image as a data sample



This is a 300×200 image

we can look at it as a 60000-dimensional sample x_n

Formulating the Problem

? *How many samples then we need?*

! *Obviously large enough to build sample covariance Σ*

Recall that the sample covariance is of the size $\Sigma \in \mathbb{R}^{D \times D}$

? *This would be a huge matrix to decompose!*

We can alternatively look at the image as

$\mathbf{X} \in \mathbb{R}^{300 \times 200}$ *and treat each column as a single 300-dimensional sample*

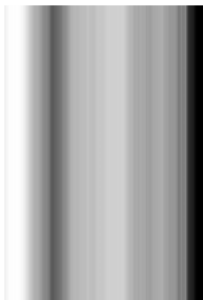
So, the image itself is a dataset!

$$\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_{200}]$$

Using PCA to Compress Images

? *How well we can compress?*

! *Let's look at the case with $K = 0$!*



Using PCA to Compress Images

? *How well we can compress?*

! *Let's look at the case with $K = 10$!*



Using PCA to Compress Images

? *How well we can compress?*

! *Let's look at the case with $K = 20$!*



Using PCA to Compress Images

- ? *How well we can compress?*
- ! *Let's look at the case with $K = 30$!*



Using PCA to Compress Images

? *How well we can compress?*

! *Let's look at the case with $K = 40$!*



Using PCA to Compress Images

? *How well we can compress?*

! *Let's look at the case with $K = 50$!*



Sounds to be good enough!

Further Read

- Bishop

- ↳ Chapter 12: *Section 12.1*

PCA

- ESL

- ↳ Chapter 14: *Section 14.1*

Principal Components

- Goodfellow

- ↳ Chapter 2

Review on Linear Algebra

- ↳ Chapter 5: *Section 5.8.1*

PCA

- MacKay

- ↳ Chapter 34

Latent Space Design